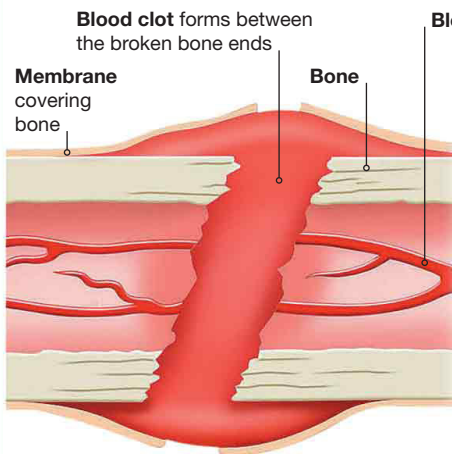


Healing fractures

Bones are strong, but sometimes they fracture or break. When this happens, a self-repair mechanism springs into action. This process may need outside help from doctors to ensure that the bones—especially the arm and leg bones—are kept straight while they heal.

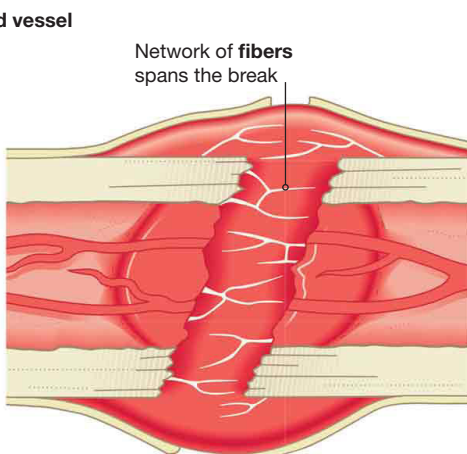
Healing in progress

This step-by-step sequence shows how a long bone heals. When a bone fractures, the body's immediate response is to stop bleeding from the bone's blood vessels. In the days and weeks that follow, new tissues are laid down that reconnect the broken bone ends. A broken bone can take months to heal completely.



IMMEDIATE RESPONSE

Blood leaking from torn blood vessels forms a jellylike clot. This seals off the vessels and stops blood from pouring into the wound.



THREE DAYS LATER

Repair cells called fibroblasts move to the fracture and produce fibers made of collagen (a protein that works as a building material). These fibers connect the broken bone ends.